

Magnetic Moments

Timothy Lim

February 19, 2026

Introduction and Theory

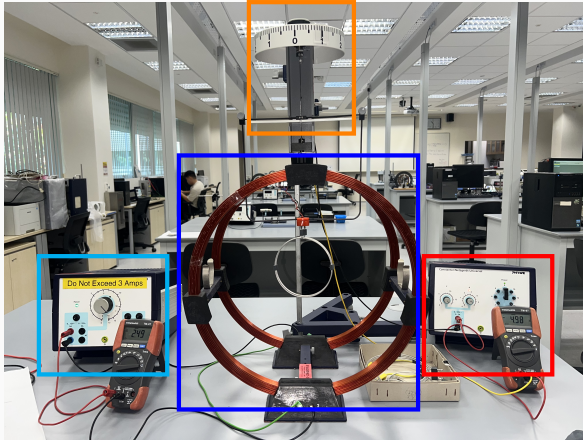
$$\vec{T} = \vec{m} \times \vec{B} \quad (1)$$

$$\vec{m} = I \cdot n \cdot \vec{A} \quad (2)$$

$$B = \frac{8\mu_0 N I_H}{5\sqrt{5}R} = c I_H \quad (3)$$

$$|\vec{T}| = c \cdot I_H \cdot n \cdot I_L \cdot |\vec{A}| \cdot \sin \alpha \quad (4)$$

Experimental Setup



Key Components:

- Helmholtz Coil Power Supply
- Helmholtz Coils, Wire Loop
- Torsion Dynamometer
- Wire Loop Power Supply

Data Analysis & Experimental Results

Data Analysis Methodology

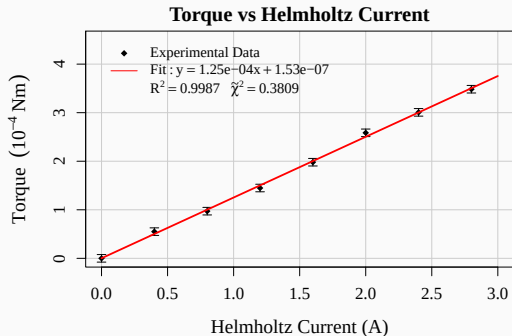
- **Linearization:** All physical equations were rearranged into the linear form with two fitting parameters (a,b):

$$y = ax + b$$

- **Regression Model:** Least Squares (Ex 1,2,4,5) and Weighted Least Squares (Ex 3)
- **Quality of Fit:** R^2 (strength of linear relationship) and $\tilde{\chi}^2$ (goodness of fit)

Ex. 1 & 2: Varying Helmholtz Current & Coil Turns

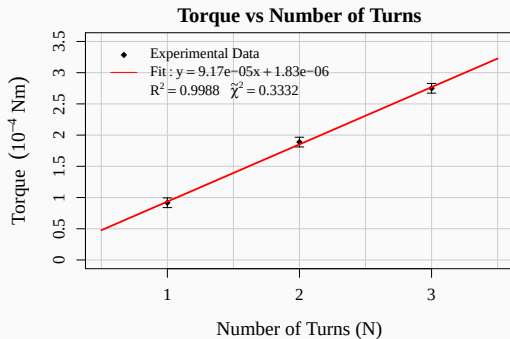
Ex. 1: Vary Helmholtz Current



$$a = (1.25 \pm 0.03) \times 10^{-4} \text{ Nm/A}$$

$$b = (0 \pm 5) \times 10^{-6} \text{ Nm}$$

Ex. 2: Vary Number of Coils (n)

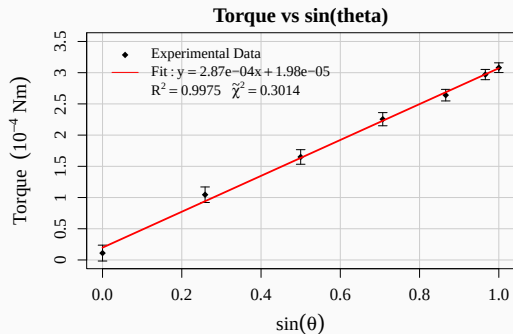


$$a = (9.2 \pm 0.5) \times 10^{-5} \text{ Nm}$$

$$b = (0.2 \pm 1.2) \times 10^{-5} \text{ Nm}$$

Ex. 3 & 4: Varying Angle & Loop Current

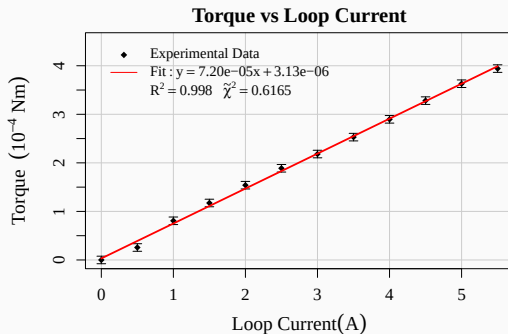
Ex. 3: Vary Angle (θ)



$$a = (2.87 \pm 0.12) \times 10^{-4} \text{ Nm}$$

$$b = (2.0 \pm 0.9) \times 10^{-5} \text{ Nm}$$

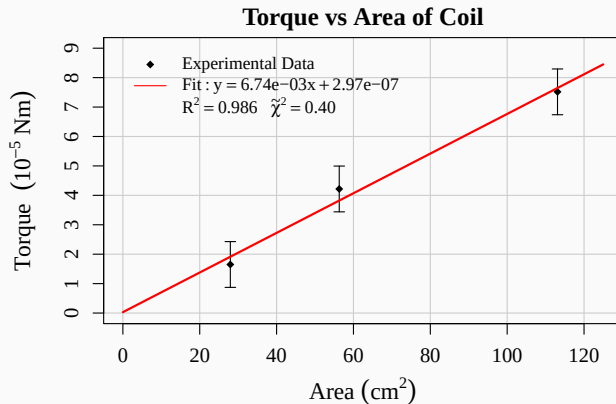
Ex. 4: Vary Loop Current (I_L)



$$a = (7.20 \pm 0.13) \times 10^{-5} \text{ Nm/A}$$

$$b = (3 \pm 4) \times 10^{-6} \text{ Nm}$$

Ex. 5: Vary Loop Diameter, Measure Torque



$$a = (6.7 \pm 1.3) \times 10^{-3} \text{ N/m}$$

$$b = (0 \pm 9) \times 10^{-6} \text{ Nm}$$

Analysis of Systematic Errors (Y-Intercepts)

Variable	Intercept (Nm)	Deviation (σ)
1. Current (I_H)	$(0 \pm 5) \times 10^{-6}$	0.03σ
2. Turns (n)	$(0.2 \pm 1.2) \times 10^{-6}$	0.15σ
3. Angle ($\sin \alpha$)	$(2.0 \pm 0.9) \times 10^{-5}$	2.12σ
4. Loop (I_L)	$(3 \pm 4) \times 10^{-6}$	0.74σ
5. Area (A)	$(0 \pm 9) \times 10^{-6}$	0.03σ

- Probability of a 2.12σ result in 5 runs $\approx 14.8\%$

Determination of Helmholtz Constant (c)

$$|\vec{T}| = c \cdot I_H \cdot n \cdot I_L \cdot |\vec{A}| \cdot \sin \alpha$$

$$a = \text{Gradient} \quad \Rightarrow \quad c_{\text{exp}} = \frac{a}{(\text{Product of Fixed Constants})}$$

Variable	c_{exp} (T/A)	
1. Current (I_H)	$(6.6 \pm 0.2) \times 10^{-4}$	
2. Turns (n)	$(6.5 \pm 0.4) \times 10^{-4}$	
3. Angle ($\sin \alpha$)	$(6.8 \pm 0.3) \times 10^{-4}$	
4. Loop (I_L)	$(8.5 \pm 0.2) \times 10^{-4}$	$\rightarrow 7.44\sigma$ difference
5. Area (A)	$(5.4 \pm 1.0) \times 10^{-4}$	

Global Analysis

1. The data has been shown to be of high statistical quality (R^2 , $\tilde{\chi}^2$, and anomalies have been accounted for)
2. Experimental values of c can now be combined using a weighted average

$$\bar{c} = (6.61 \pm 0.15) \times 10^{-4} T/A$$

$$\text{Literature: } c = 6.92 \times 10^{-4} T/A$$

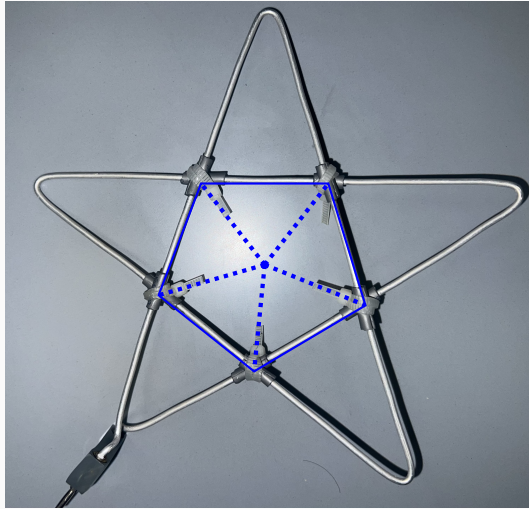
$$\bar{\mu}_0 = \frac{c \cdot 5\sqrt{5} \cdot R}{8N}$$

$$\bar{\mu}_0 = (1.20 \pm 0.03) \times 10^{-6} H/m$$

Literature: $\mu_0 = 1.257 \times 10^{-6} H/m$

Statistical Deviation: 4.5% or 1.89σ

Star-Shaped Wire: Area Calculation



Star-Shaped Wire

- **Geometric Area:** $(81.8 \pm 2.5) \text{ cm}^2$
- **Magnetic Area:** $(121 \pm 9) \text{ cm}^2$
- **Predicted Magnetic Area (2 Pentagons):** $(113 \pm 3) \text{ cm}^2$
- **Statistical Difference:** 7.0% or 0.84σ

Discussion

- **Fundamental Law:** We successfully validated the vector relationship $\vec{T} = \vec{m} \times \vec{B}$ by confirming the scalar dependencies in the experimental model:

$$|\vec{T}| = c \cdot I_H \cdot n \cdot I_L \cdot |\vec{A}| \cdot \sin \alpha$$

- **Global Consistency:** Agreement with 4/5 experiments, good R^2 and $\tilde{\chi}^2$ values
- **Physical Constants:** c and μ_0 statistically consistent with literature
- **Geometric Verification:** Predicted magnetic area of star

Error Analysis

1. **Torque:** Instrument uncertainty and alignment ambiguity

$$\delta\tau = \sqrt{0.05^2 + 0.05^2} \times (\text{Scaling Factors}) \approx 7.78 \times 10^{-6} \text{ Nm}$$

2. **Angle (Ex.3):** Instrument uncertainty (2°) with error propagation

$$\delta \sin \theta = |\cos \theta| \times \delta \theta$$

$$\delta\tau_3 = \sqrt{\delta\tau^2 + (\text{gradient} \times \delta \sin \theta)^2}$$

Error Analysis

Ex.4 (high value):

- Alignment of Helmholtz coils and Wire Loop
- Thermal softening of torsion dynamometer

Ex.5 (low value):

- Small data set $\rightarrow$ high uncertainty
- Overestimation of loop area
- α not perfectly aligned at 90°

Improvements: Quantify positional sensitivity, calibration for torsion constant (heating)

Thank you!
